



CRYPTO REALM GAMEFI ECONOMY PROTOCOL

A Production-Grade GameFi Economy on Base Sepolia

BLOCKCHAIN TECHNOLOGIES 2 — FINAL PROJECT

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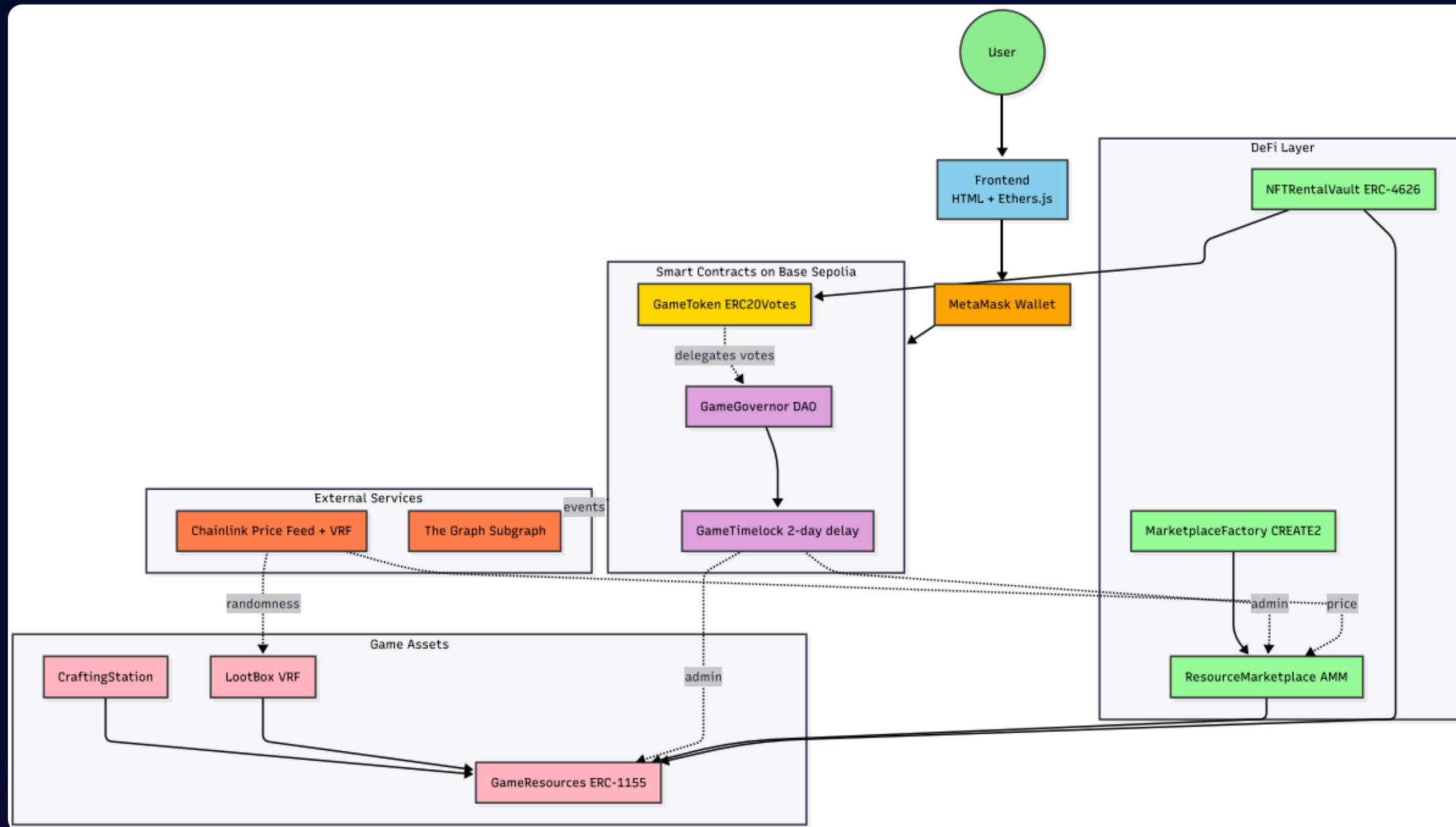


WHAT IS CRYPTO REALM?

A complete on-chain game economy on Base Sepolia
Players collect resources, craft items, and trade on an AMM
Items can be rented from an ERC-4626 collateralized vault
Chainlink VRF powers randomized loot box drops
DAO governance with two-day timelock controls all parameters



SYSTEM ARCHITECTURE



- Twelve production contracts organized into seven domain folders
- OpenZeppelin v5 as the foundation
- Chainlink integrations for price feeds and verifiable randomness
- The Graph subgraph indexes events into four entities



WHO BUILT WHAT

Bigali (Assets + Vault)

GameResources ERC-1155
CraftingStation recipe engine
NFTRentalVault ERC-4626



Asylbek (Marketplace +
Upgradeability)

ResourceMarketplace AMM
MarketplaceV1 / V2 UUPS
MarketplaceFactory CREATE2
MarketplaceMath Yul library

Miras (Governance + Oracles)

GameToken ERC-20 Votes
GameTimelock + GameGovernor
PriceOracleAdapter Chainlink
LootBox VRF + Subgraph



12 PRODUCTION CONTRACTS

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- All contracts compile cleanly with Solidity 0.8.24 and Cancun EVM
- Strict via-IR optimization enabled
- 47 distinct custom errors instead of require strings
- Every privileged function gated by AccessControl roles

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```
user@WIN-ON5SS8V15M6 MINGW64 ~/Desktop/Blockchain2/final/defi-superapp-day1/defi-superapp (main)
• $ forge build
Warning: This is a nightly build of Foundry. It is recommended to use the latest stable version. To mute this warning set

[.] Compiling...
[.] Compiling 18 files with Solc 0.8.24
[.] Solc 0.8.24 finished in 144.26s
Compiler run successful!
```


88 TESTS + PASSING

- 57 unit tests covering every external function and revert path
- 12 fuzz tests with 256 random inputs each
- 5 invariant tests with 128,000 randomized calls
- 3 fork tests against the live Base Sepolia Chainlink feed
- 5 vulnerability case studies with before-and-after fixes

```
$ forge test --summary
Warning: This is a nightly build of Foundry. It is recommended to use the latest stable version

[ ] Compiling...
No files changed, compilation skipped

Ran 3 tests for test/vulnerabilities/AccessControlCaseStudy.t.sol:AccessControlCaseStudyTest
[PASS] test_After_FixedAdmin_AccessControlBlocksProxyAttack() (gas: 24493)
[PASS] test_After_FixedAdmin_LegitimateAdminCanSetValue() (gas: 36225)
[PASS] test_Before_VulnerableAdmin_TxOriginAllowsProxyAttack() (gas: 43769)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 1.79ms (483.30µs CPU time)

Ran 6 tests for test/unit/MarketplaceFactory.t.sol:MarketplaceFactoryTest
[PASS] test_DeployWithCreate2_DifferentSaltsProduceDifferentAddresses() (gas: 2326006)
[PASS] test_DeployWithCreate2_MatchesPredictedAddress() (gas: 1196579)
[PASS] test_DeployWithCreate2_RevertWhen_CallerUnauthorized() (gas: 11495)
[PASS] test_DeployWithCreate_AssignsValidAddress() (gas: 1187537)
[PASS] test_DeployWithCreate_RevertWhen_AlreadyExists() (gas: 1187499)
[PASS] test_DeployWithCreate_RevertWhen_CallerUnauthorized() (gas: 10959)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 5.46ms (4.08ms CPU time)

Ran 10 tests for test/unit/PriceOracleAdapter.t.sol:PriceOracleAdapterTest
[PASS] test_GetLatestPrice_RevertWhen_DataIsStale() (gas: 51116)
[PASS] test_GetLatestPrice_RevertWhen_FeedNotConfigured() (gas: 10670)
[PASS] test_GetLatestPrice_RevertWhen_PriceIsNegative() (gas: 55133)
[PASS] test_GetLatestPrice_RevertWhen_PriceIsZeroOrNegative() (gas: 50489)
[PASS] test_GetLatestPrice_ScalesEightDecimalsToEighteen() (gas: 52857)
[PASS] test_PriceUpdate_AfterStalenessReverts_FreshDataPasses() (gas: 63309)
[PASS] test_RegisterPriceFeed_RevertWhen_CallerUnauthorized() (gas: 13729)
[PASS] test_RegisterPriceFeed_StoresFeedAddress() (gas: 38243)
[PASS] test_SetMaximumStaleness_RevertWhen_CallerUnauthorized() (gas: 11625)
[PASS] test_SetMaximumStaleness_UpdatesValue() (gas: 17981)
Suite result: ok. 10 passed; 0 failed; 0 skipped; finished in 4.60ms (2.97ms CPU time)

Ran 5 tests for test/unit/ResourceMarketplace.t.sol:ResourceMarketplaceTest
[PASS] test_AddLiquidity_FirstProviderReceivesSquareRootMinusMinimum() (gas: 206370)
[PASS] test_AddLiquidity_RevertWhen_DeadlineExpired() (gas: 17579)
[PASS] test_RemoveLiquidity_ReturnsProportionalAmounts() (gas: 204956)
[PASS] test_Swap_OutputAmountMatchesConstantProductFormula() (gas: 238156)
[PASS] test_Swap_RevertWhen_PoolDoesNotExist() (gas: 19303)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 6.57ms (3.86ms CPU time)

Ran 6 tests for test/unit/MarketplaceUpgrade.t.sol:MarketplaceUpgradeTest
[PASS] test_GetContractVersion_ReturnsV1() (gas: 14990)
[PASS] test_Initialize_SetsInitialState() (gas: 18814)
[PASS] test_SetProtocolFee_OnlyOwnerCanUpdate() (gas: 22933)
[PASS] test_SetProtocolFee_RevertWhen_CallerNotOwner() (gas: 15950)
[PASS] test_UpgradeToV2_PreservesStateAndAddsFunctionality() (gas: 681735)
[PASS] test_UpgradeToV2_RevertWhen_CallerNotOwner() (gas: 636110)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 4.47ms (2.75ms CPU time)

Ran 7 tests for test/unit/NFTRentalVault.t.sol:NFTRentalVaultTest
[PASS] test_Constructor_SetsAssetAndResources() (gas: 11538)
[PASS] test_CreateRental_RevertWhen_CallerNotAdmin() (gas: 12089)
[PASS] test_CreateRental_RevertWhen_ZeroCollateral() (gas: 17317)
[PASS] test_CreateRental_TransfersCollateralAndItem() (gas: 279290)
[PASS] test_EndRental_ReturnsCollateralAfterExpiration() (gas: 244443)
[PASS] test_EndRental_RevertWhen_AlreadyEnded() (gas: 247100)
[PASS] test_EndRental_RevertWhen_NotExpired() (gas: 280128)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 10.12ms (6.13ms CPU time)

Ran 2 tests for test/vulnerabilities/ReentrancyCaseStudy.t.sol:ReentrancyCaseStudyTest
[PASS] test_After_FixedBank_AttackerCannotDrain() (gas: 196976)
[PASS] test_Before_VulnerableBank_AttackerDrainsBeyondDeposit() (gas: 255207)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 4.98ms (1.42ms CPU time)

Ran 9 tests for test/unit/LootBox.t.sol:LootBoxTest
[PASS] test_Constructor_SetsInitialState() (gas: 14074)
[PASS] test_FulfillRandomWords_RevertWhen_NotCoordinator() (gas: 9437)
[PASS] test_OpenLootBox_BurnsWoodAndCreatesRequest() (gas: 91578)
[PASS] test_OpenLootBox_FulfilledImmediately_AwardsCommonResource() (gas: 111793)
[PASS] test_OpenLootBox_FulfilledWithRareRoll_AwardsRareResource() (gas: 142105)
[PASS] test_SetLootBoxCost_AdminCanModify() (gas: 16947)
[PASS] test_UpdateDropProbabilities_AdminCanModify() (gas: 44146)
[PASS] test_UpdateDropProbabilities_RevertWhen_AllZeroWeights() (gas: 13237)
[PASS] test_UpdateDropProbabilities_RevertWhen_CallerUnauthorized() (gas: 12236)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 8.21ms (5.81ms CPU time)
```

```
Ran 7 tests for test/unit/GameResources.t.sol:GameResourcesTest
[PASS] test_BurnResource_ReducesBalance() (gas: 54846)
[PASS] test_MintResource_AssignsCorrectBalance() (gas: 49646)
[PASS] test_MintResource_RevertWhen_CallerLacksMinterRole() (gas: 16414)
[PASS] test_MintResource_RevertWhen_InvalidResourceId() (gas: 18047)
[PASS] test_MintResource_RevertWhen_ZeroAmount() (gas: 18797)
[PASS] test_Pause_BlocksMinting() (gas: 41265)
[PASS] test_Unpause_RestoresMinting() (gas: 57771)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 7.79ms (5.19ms CPU time)
```

```
Ran 6 tests for test/unit/GameToken.t.sol:GameTokenTest
[PASS] test_Constructor_SetsMetadata() (gas: 18321)
[PASS] test_Delegate_VotingPowerActivates() (gas: 180112)
[PASS] test_MintTokens_AdminCanMint() (gas: 108937)
[PASS] test_MintTokens_RevertWhen_CallerLacksRole() (gas: 11929)
[PASS] test_MintTokens_RevertWhen_ExceedsMaxSupply() (gas: 13961)
[PASS] test_Permit_NonceStartsAtZero() (gas: 8171)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 7.33ms (2.70ms CPU time)
```

```
Ran 6 tests for test/fuzz/GovernanceFuzz.t.sol:GovernanceFuzzTest
[PASS] testFuzz_Delegate_VotingPowerMatchesBalance(address,uint256) (runs: 256, µ: 181216, ~: 181114)
[PASS] testFuzz_Mint_BalanceIncreasesByMintedAmount(address,uint256) (runs: 256, µ: 110243, ~: 110141)
[PASS] testFuzz_Mint_RevertWhen_CallerLacksMinterRole(address,uint256) (runs: 256, µ: 12656, ~: 12554)
[PASS] testFuzz_Mint_RevertWhen_ExceedsMaxSupply(uint256) (runs: 256, µ: 14287, ~: 14327)
[PASS] testFuzz_Transfer_PreservesTotalSupply(address,address,uint256) (runs: 256, µ: 121717, ~: 121606)
[PASS] testFuzz_VotingPower_PreservedAfterTransferIfBothDelegated(uint256) (runs: 256, µ: 285505, ~: 285401)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 104.79ms (326.93ms CPU time)
```

```
Ran 4 tests for test/fuzz/GameResourcesFuzz.t.sol:GameResourcesFuzzTest
[PASS] testFuzz_BatchMint_AllResourcesAssigned(uint256,uint256,uint256) (runs: 256, µ: 100401, ~: 100315)
[PASS] testFuzz_Burn_BalanceDecreasesByAmount(uint256,uint256) (runs: 256, µ: 50059, ~: 50043)
[PASS] testFuzz_Mint_BalanceMatchesAmount(address,uint256) (runs: 256, µ: 46219, ~: 46131)
[PASS] testFuzz_Mint_RevertWhen_InvalidResourceId(uint256) (runs: 256, µ: 14624, ~: 14642)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 103.90ms (195.96ms CPU time)
```

```
Ran 4 tests for test/fuzz/ResourceMarketplaceFuzz.t.sol:ResourceMarketplaceFuzzTest
[PASS] testFuzz_AddLiquidity_FirstDepositMintsExpectedShares(uint256,uint256) (runs: 256, µ: 209382, ~: 210175)
[PASS] testFuzz_RemoveLiquidity_ReturnsNoMoreThanDeposited(uint256) (runs: 256, µ: 207006, ~: 207924)
[PASS] testFuzz_Swap_KInvariantNeverDecreases(uint256) (runs: 256, µ: 240658, ~: 240722)
[PASS] testFuzz_Swap_OutputAlwaysLessThanReserve(uint256) (runs: 256, µ: 239027, ~: 239091)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 103.54ms (351.59ms CPU time)
```

```
Ran 7 tests for test/unit/MarketplaceMath.t.sol:MarketplaceMathTest
[PASS] testFuzz_BothImplementations_AlwaysEqual(uint256,uint256,uint256) (runs: 256, µ: 1888, ~: 1765)
[PASS] testFuzz_OutputAlwaysLessThanOutputReserve(uint256,uint256,uint256) (runs: 256, µ: 1654, ~: 1531)
[PASS] test_BothImplementations_ProduceIdenticalResults() (gas: 155)
[PASS] test_GetAmountOutPureSolvency_BasicCalculation() (gas: 331)
[PASS] test_GetAmountOutYulOptimized_BasicCalculation() (gas: 507)
[PASS] test_LargeAmounts_ProduceIdenticalResults() (gas: 485)
[PASS] test_SmallAmounts_ProduceIdenticalResults() (gas: 243)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 115.89ms (222.31ms CPU time)
```

```
Ran 3 tests for test/fork/PriceOracleFork.t.sol:PriceOracleForkTest
[PASS] test_Fork_ChainlinkFeedHasCorrectDecimals() (gas: 8949)
[PASS] test_Fork_ChainlinkFeedReturnsPrice() (gas: 58571)
[PASS] test_Fork_ChainlinkFeedReturnsRecentData() (gas: 19092)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 3.96s (5.94s CPU time)
```

```
Ran 5 tests for test/unit/Governance.t.sol:GovernanceTest
[PASS] test_FullProposalLifecycle_ProposeVoteQueueExecute() (gas: 404233)
[PASS] test_GovernorParameters_MatchSpecification() (gas: 17600)
[PASS] test_Mint_RevertWhen_ExceedsMaximumSupply() (gas: 14044)
[PASS] test_Token_HasCorrectMetadata() (gas: 18417)
[PASS] test_VotingPower_MatchesDelegatedBalance() (gas: 16045)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 10.41ms (5.99ms CPU time)
```



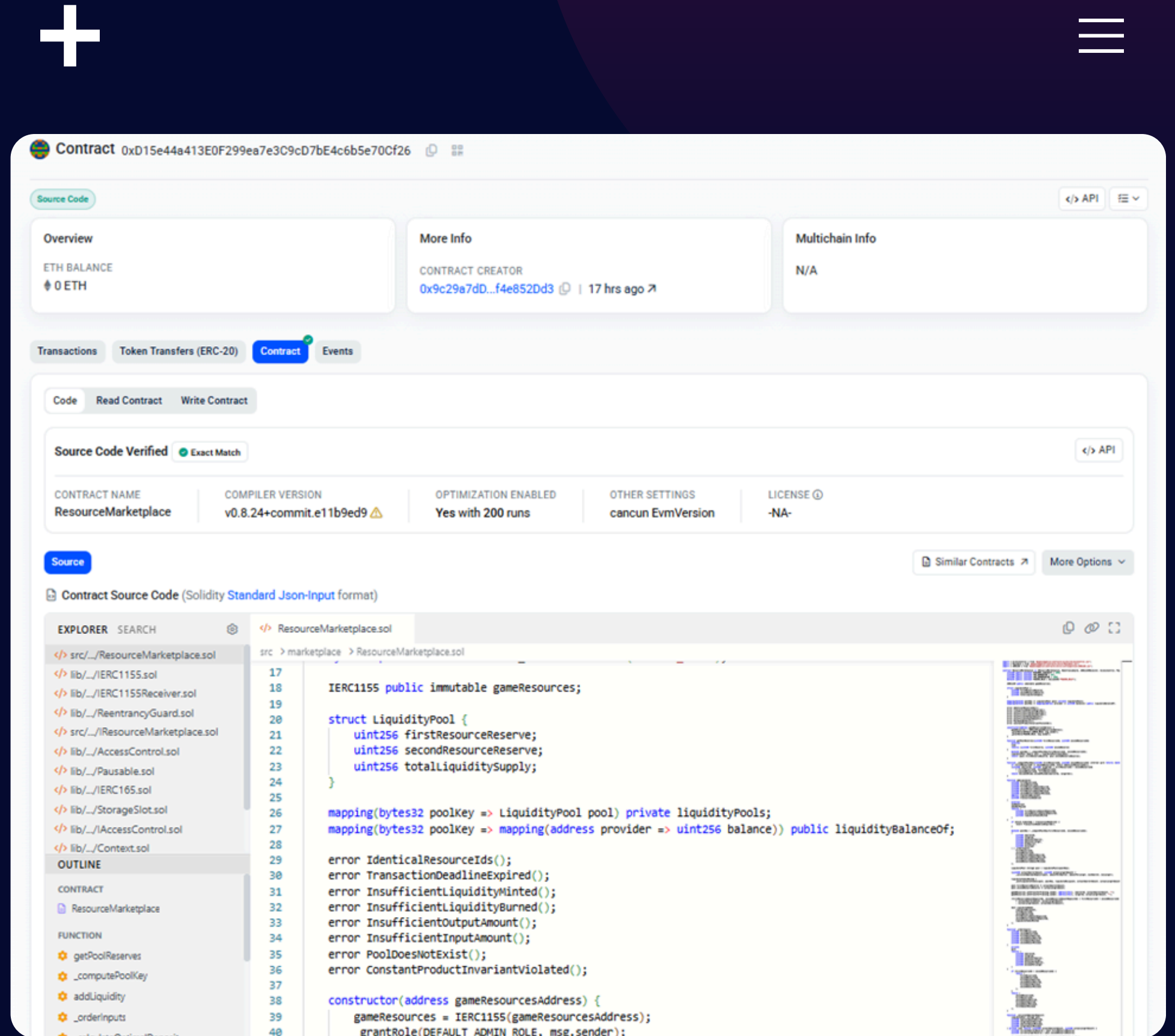

SECURITY

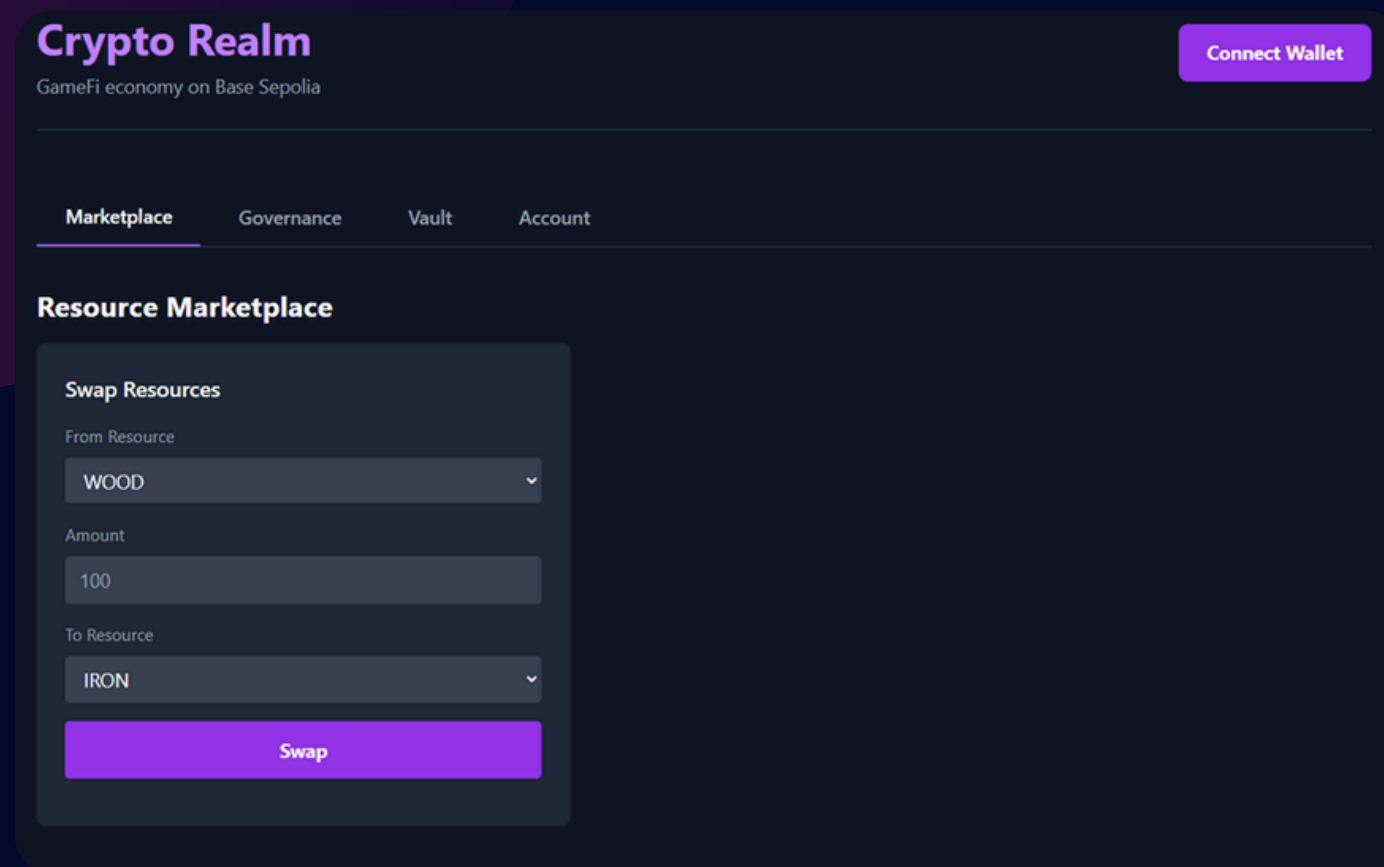
- Slither: zero High, zero Medium findings
- Checks-Effects-Interactions across all mutating functions
- AccessControl, Pausable, ReentrancyGuard, SafeERC20 throughout
- Internal audit report: 15 pages, 8 documented Low findings
- Two reproduced-and-fixed vulnerabilities: reentrancy and tx.origin



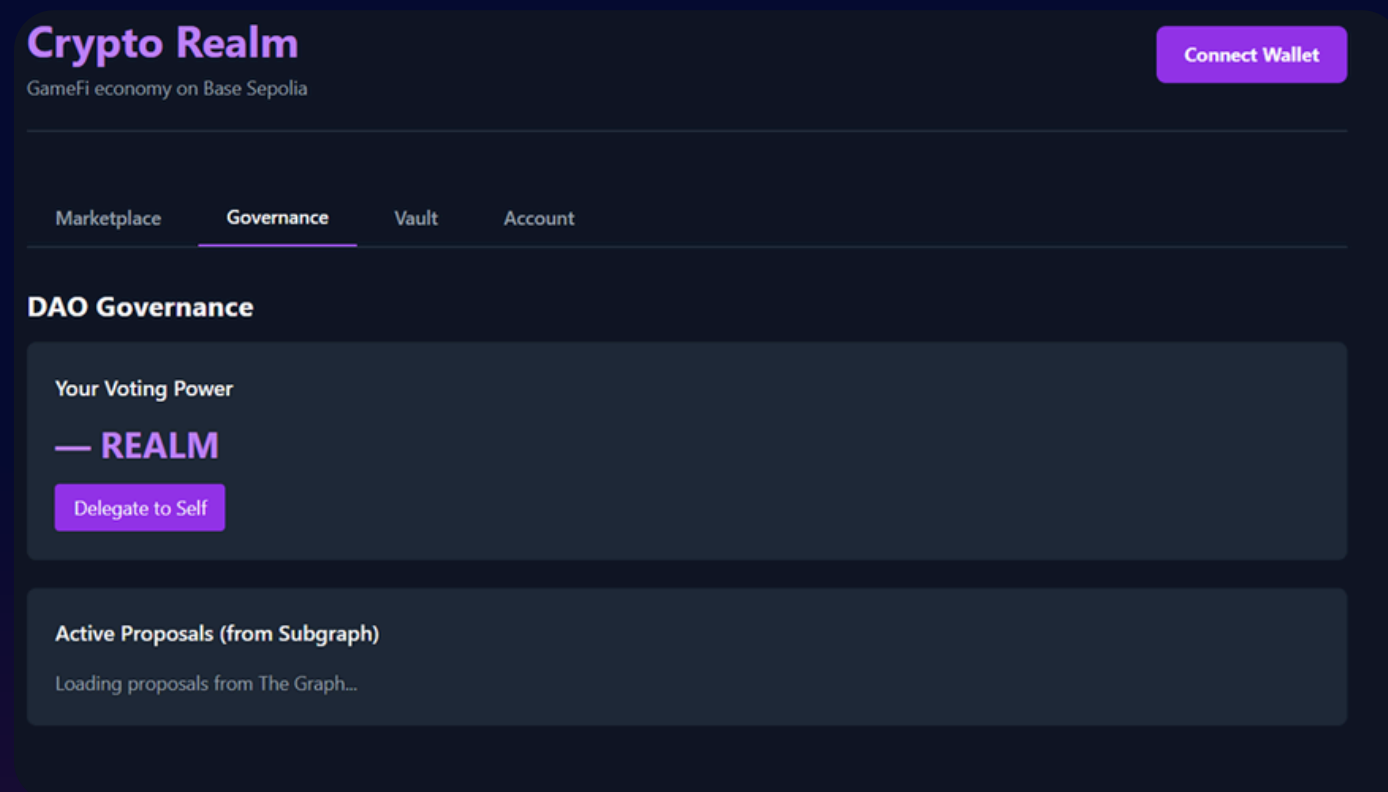
DEPLOYED AND VERIFIED ON BASE SEPOLIA

- 13 contracts deployed (12 production + 1 ERC1967 proxy)
- 13 of 13 verified on BaseScan with source code
- Post-deployment verification script: 12 of 12 checks pass
- Admin role transferred to GameTimelock after deployment





DAPP FRONTEND



- HTML and Ethers.js v5, styled with Tailwind CSS
- MetaMask wallet integration
- Network detection with one-click Base Sepolia switch
- Three state-changing transactions: swap, delegate, deposit, vote
- Active proposal list with three vote options

GAS OPTIMIZATION RESULTS

- Yul-optimized AMM math: 25% gas savings over pure Solidity
- Immutable variables eliminate SLOAD (2100 gas to 3 gas per read)
- Custom errors save 50+ gas per revert
- L1 versus L2: 99.8% cost reduction on Base versus Ethereum mainnet

```
You, 8 hours ago | 1 author (You)
1 Warning: This is a nightly build of Foundry. It is recommended to use the latest stable version.
2
3 Compiling 17 files with Solc 0.8.24
4 Solc 0.8.24 finished in 181.78s
5 Compiler run successful!
6
7 Ran 6 tests for test/unit/MarketplaceFactory.t.sol:MarketplaceFactoryTest
8 [PASS] test_DeployWithCreate2_DifferentSaltsProduceDifferentAddresses() (gas: 2374682)
9 [PASS] test_DeployWithCreate2_MatchesPredictedAddress() (gas: 1225023)
10 [PASS] test_DeployWithCreate2_RevertWhen_CallerUnauthorized() (gas: 33439)
11 [PASS] test_DeployWithCreate_AssignsValidAddress() (gas: 1215469)
12 [PASS] test_DeployWithCreate_RevertWhen_AlreadyExists() (gas: 1232363)
13 [PASS] test_DeployWithCreate_RevertWhen_CallerUnauthorized() (gas: 32391)
14 Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 17.45ms (8.46ms CPU time)
15
16 Ran 6 tests for test/unit/CraftingStation.t.sol:CraftingStationTest
17 [PASS] test_CraftRecipe_ConsumesInputsAndProducesOutput() (gas: 375663)
18 [PASS] test_CraftRecipe_RevertWhen_RecipeDoesNotExist() (gas: 50542)
19 [PASS] test_CraftRecipe_RevertWhen_RecipeNotActive() (gas: 279208)
20 [PASS] test_RegisterRecipe_RevertWhen_CallerUnauthorized() (gas: 33442)
21 [PASS] test_RegisterRecipe_StoresRecipeData() (gas: 212191)
22 [PASS] test_SetRecipeActiveStatus_AdminCanToggle() (gas: 251907)
23 Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 20.29ms (5.42ms CPU time)
24
```



LIVE DEMO

- Open the dApp in a browser
- Connect MetaMask to Base Sepolia
- Execute a token swap on the marketplace
- View the transaction on BaseScan





SUMMARY

- 12 production contracts, 88 tests, 0 High/Medium findings
- 13 of 13 contracts verified on Base Sepolia
- 15-page audit report, 13-page architecture document
- Repository: github.com/bgl96395/Blockchain2_Project

